
DATADECEPTION

A CITATION-FIRST, SCHEMA-VALIDATED DATASET OF MANUFACTURER-CLAIM VS.
INDEPENDENT-MEASUREMENT DIVERGENCE IN THE U.S. ELECTRIC-VEHICLE MARKET

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ABSTRACT

Tesla advertises 318 mi of range for the Cybertruck Dual Motor. Independent measurement at a constant 70 mph delivers 224 mi—a 30% shortfall, characterized by MotorTrend as “among the worst performers in our database of 56 EVs” [22]. Such gaps are not random. We introduce **DataDeception**, a structured, citation-first dataset and methodology for systematically documenting where manufacturer claims about consumer products diverge from independently-measured reality. Existing fact-checking systems (browser extensions, AI verifiers, community labelers) target news and social-media misinformation; almost none address *commercial spec-sheet deception*, where the harm vector is consumer purchase decisions and the legal frame (FTC §5, comparative-advertising statutes) is well established. The DataDeception methodology decomposes each product into a tuple of (*manufacturer, model, model year, trim*), attaches one or more *deception records* per relevant axis, links each record to a curated *tactic* (a recurring deception pattern), and grades evidence A–D against a published rubric. We instantiate the methodology in the U.S. electric-vehicle vertical, releasing v0.1: **13 products × 28 records × 13 tactics across 8 manufacturers**, all validated against JSON Schema 2020-12. Preliminary findings on this non-randomized sample show systematic divergence between EPA combined-cycle range and independently-measured highway range (mean gap −15%, range −30% to +3%), with the `epa_vs_highway_70mph` tactic recurring across ≥ 8 manufacturers including counter-examples in both directions. We deliberately include counter-examples (Mercedes EQS, Ford Mach-E) to address confirmation-bias concerns inherent in this kind of work. Building on the dataset-documentation tradition of Gebru et al. [8] and the empirical-taxonomy approach of Mathur et al. [12], the dataset, schema, validators, and figure-generation pipeline are released under MIT (code) and CC-BY-4.0 (data).

Keywords: consumer protection · deceptive advertising · electric vehicles · open data · methodology · taxonomy · falsifiability

1 Introduction

Most modern claim-vs-reality projects target news and political misinformation: AI fact-checking browser extensions, community-note systems, and crowd-source labelers [17]. The methodological consensus has converged: warning labels work, even for distrustful audiences, but the *trust-collapse* dynamic limits ceiling effectiveness. Less attention has been paid to a structurally different problem: *commercial-claims deception*, where a manufacturer prominently advertises a quantitative spec (range, charging speed, peak power, towing capacity, autonomy level) that a buyer relies upon for a major purchase decision and that diverges from what independent measurement reveals.

The commercial-claims problem differs from the political-misinformation problem in three important ways. First, the truth has a denominator: “up to 500 miles of range” is a precise testable assertion, not a contested narrative. Second, the legal framework already exists: FTC §5 covers deceptive acts in commerce, decades of comparative-advertising case law (e.g. *Kraft, Inc.* [7]; *Intuit v. FTC* [1]) define what counts as a deceptive claim. Third, the buyer’s identity is not at stake: showing a buyer that their preferred vehicle’s range is overstated does not implicate their tribal affiliation, and so does not trigger the backfire effects that limit political fact-checking.

We argue that the bottleneck in commercial-claims accountability is *not* a methodological one (the rules of fair claims are well understood) but a *data-availability* one: independent measurements exist (Consumer Reports, MotorTrend, Edmunds, Out of Spec, Recharged), but they are scattered across publications, often paywalled, and not structured for cross-product analysis. The recent paywalling of RTINGS test data [21], an industry-leading independent measurement source, illustrates this fragility.

This paper introduces **DataDeception**, a methodology and an instantiating dataset, with three contributions:

1. A **schema** for representing manufacturer claims and independent-measurement reality as falsifiable tuples, with mandatory citation on both sides and explicit handling of categorical (non-numeric) deceptions (§4.1).
2. A **tactic taxonomy** that lifts individual claim/reality records to the level of theoretical patterns. The taxonomy covers 13 tactics across 5 families and is open to extension (§4).
3. An **instantiation** in the U.S. EV market with 13 products and 28 records, validated against JSON Schema 2020-12 with referential-integrity checks. Tier-1 NHTSA recall data is integrated. We deliberately include counter-examples to mitigate confirmation bias (§5).

This work builds on the dataset-documentation tradition initiated by Gebru et al. [8] (and refined by Mitchell et al. [14] and Pushkarna et al. [18]), and the empirical-taxonomy methodology demonstrated for deceptive design by Gray et al. [9] and Mathur et al. [12, 13]. What is novel here is the application of these traditions to *manufacturer-claim deception* with mandatory citations on both the claim and the reality side, an evidence rubric calibrated to consumer-protection law, and a vertical (consumer EVs) where the artifacts are physical and the gap is measurable in absolute units. The dataset is released under CC-BY-4.0 with code under MIT. A consumer-facing surface auto-generated from the same JSON is published alongside the technical artifact.

Findings at a glance

–30%. Tesla Cybertruck Dual Motor advertises 318 mi range; instrumented test at 70 mph delivers 224 mi [22]. The Range Extender promised in 2019 to bridge the 500 mi gap was cancelled in May 2025 with full refunds, never delivered to a single customer [20].

8 manufacturers. The `epa_vs_highway_70mph` tactic recurs across Tesla, Ford, Rivian, Hyundai, Lucid, Mercedes-Benz, GMC, and Volkswagen, decisively exceeding our pre-registered ≥ 3 threshold for non-idiosyncratic patterning.

9 recalls. Rivian R1S 2025 carries 9 active NHTSA recall campaigns; Hyundai Ioniq 5 2025 carries 8, including two separate high-voltage traction-battery campaigns. Brand quality language does not, in general, reflect this variance.

2.88M vehicles. NHTSA’s October 2025 “Full Self-Driving” investigation covers 2.88 M vehicles, 14 crashes, 23 injuries. A California Administrative Law Judge ruled in December 2025 that the FSD name is “unambiguously false and counterfactual” [16].

+72 mi. Counter-examples exist: Mercedes EQS exceeded its EPA estimate by 72 mi in the Edmunds EV range loop [6], and Ford Mach-E matched EPA in independent testing. We include them by design.

2 Related Work

2.1 Deceptive design taxonomies

The most direct methodological predecessors are the dark-patterns literature. Gray et al. [9] defined the original five high-level pattern families (*nagging*, *obstruction*, *sneaking*, *interface interference*, *forced action*) and remains the most-cited paper in deceptive-design research. Mathur et al. [12] performed the largest empirical audit to date—a crawl of ~11K shopping websites yielding 1,818 dark-pattern instances across 15 types and 7 categories—and is the

closest methodological analog to DataDeception in the empirical-cross-product audit dimension. Mathur, Kshirsagar, and Mayer [13] subsequently formalized the definitional question (“what makes a dark pattern dark?”) along design attributes, normative considerations, and measurement methods. Luguri and Strahilevitz [11] bridged this body of work to the legal tradition (FTC §5, UDAP statutes) through randomized experiments. Most recently, the “50 Shades” unified taxonomy [2] merges prior taxonomies and adds multimodal detection. Our 5-family tactic taxonomy (§4) is constructed in this lineage but applies the empirical-taxonomy approach to manufacturer spec-claim deception rather than UI dark patterns.

2.2 Dataset documentation and audit frameworks

The other direct lineage is dataset documentation. Gebru et al. [8] introduced “datasheets for datasets,” a lifecycle-driven documentation framework (Motivation, Composition, Collection, Preprocessing, Uses, Distribution, Maintenance) that has become the implicit standard for dataset papers in cs.LG and cs.CY. Mitchell et al. [14] and Pushkarna et al. [18] extended the framework to per-artifact and stakeholder-aware reporting. Raji et al. [19] formalized internal algorithmic auditing as an end-to-end framework. The CheckThat! shared-task series [3] establishes evaluation conventions for claim verification. Our methodology section (§3) is organized along Gebru’s lifecycle for this reason; our evidence-grade rubric (§3.3) is in the spirit of Raji’s internal-audit instrument.

2.3 Misinformation fact-checking, comparative advertising, and independent EV measurement

Misinformation fact-checking. Recent work [17] validates that warning labels reduce belief in and sharing of false news (27% and 25% reductions, respectively) even for participants distrustful of fact-checkers (12–16% reductions). The browser-extension space includes Facticity, TruthLens, FactSentinel, TruthPilot, factcheckit, Storypact, FactScope, and Webcite, plus the open-source SatyaMark and a recently-funded (€3.6M, April 2026) startup Flare. All target news and social-media veracity; we found no consumer extension targeting commercial-claim deception specifically.

Comparative advertising case law. Sectional research on FTC enforcement (*Kraft* [7], *Intuit v. FTC* [1], the Avast deceptive-privacy settlement) establishes both the regulatory framework and the long lag between first deception observation and enforcement action. The recent California DMV ruling against Tesla’s “Full Self-Driving” [16] illustrates how categorical (linguistic) overclaims are addressed within the existing legal frame.

Independent EV measurement. Consumer Reports’ real-world range program [4, 5] provides the largest single body of comparable Tier-2 data: 27 EVs tested at constant 70 mph as of April 2026. MotorTrend’s instrumented Road-Trip Range tests [22] and Edmunds’ EV range loop [6] provide independent corroboration. Crowd-sourced data exists at evrangetools.com, Recurrent Auto, and A Better Routeplanner. To our knowledge, no prior work has integrated these sources into a structured, publicly-licensed dataset with a tactic taxonomy.

2.4 Comparison to prior work

We position DataDeception by contrast with the closest predecessors:

- **vs. Mathur et al. [12] (*Dark Patterns at Scale*).** Mathur audited ~11K shopping websites for UI dark patterns at scale; the unit of deception is a UI element. We audit a much smaller cross-product set (13 EVs) but at greater per-record depth: every record carries paired claim/reality citations, a gap quantification or severity grade, and a tactic assignment. Mathur establishes that UI deception is patterned and prevalent; we extend that framing to *spec-sheet* deception with measurable physical referents.
- **vs. Gebru et al. [8] (*Datasheets for Datasets*).** Gebru documents *datasets* for ML training; we apply analogous lifecycle documentation to a *deception-record corpus* where each record itself is documentation of a manufacturer claim. Our methodology section (§3) is organized along Gebru’s lifecycle as a deliberate genre choice.
- **vs. Mathur, Kshirsagar, and Mayer [13] (*What Makes a Dark Pattern ... Dark?*).** Their framework is definitional; ours is operational. Where they ask whether a given UI pattern qualifies as deceptive, we presuppose deception and ask: *can we produce a falsifiable record of it backed by tier-classified citations on both sides?*
- **vs. Pennycook & Rand [17].** Their work shows warning labels mitigate belief in already-shared news content; we work at the upstream stage of asserting that a given commercial claim diverges from measurement. The

two complement: a structured corpus of validated divergences is a prerequisite for any commercial-claims warning-label intervention.

- **vs. Luguri & Strahilevitz [11].** They run randomized experiments on dark-pattern susceptibility and connect findings to FTC §5; we provide the structured artifact (records, tactics, gap formalism) that a comparable dataset for spec-claim deception would need before similar experiments are runnable.

To our knowledge no prior work integrates a citation-first claim/reality schema, a tactic taxonomy with cross-manufacturer corroboration thresholds, and Tier-1 regulatory open data (NHTSA recalls) into a single publicly-licensed artifact for the consumer-EV vertical.

3 Methodology

The methodology is presented along the lifecycle structure of Gebru et al. [8]: *Motivation, Composition, Collection, Preprocessing, Uses, Distribution, Maintenance*. The **deception record** is the unit of analysis: a tuple

$$\mathcal{D} = \langle \text{product, axis, sub_axis, claim, reality, tactic, gap, grade} \rangle,$$

where *claim* and *reality* each carry one or more *sources* with a verbatim quote, a four-tier classification, and (for manufacturer claims) a web.archive.org snapshot URL. Figure 1 summarizes the construction workflow.

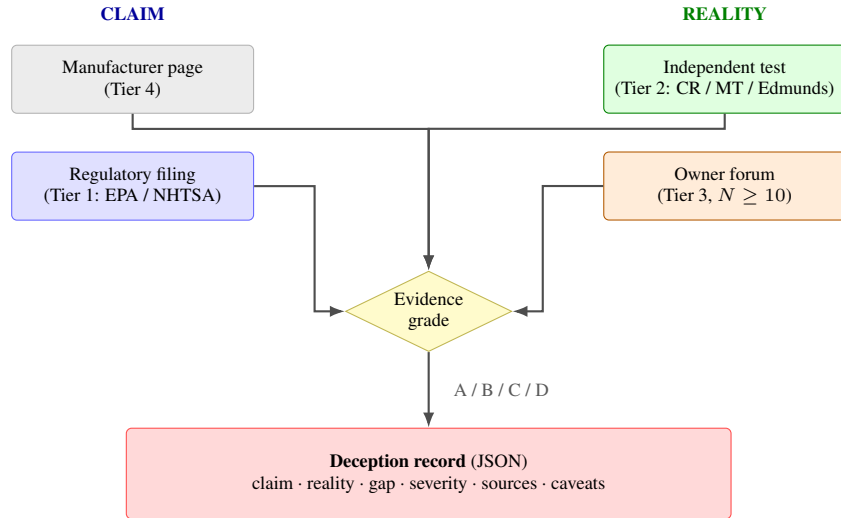


Figure 1: Methodology: how a single deception record is constructed from claim and reality sources. Tiered-source classification feeds the evidence-grade rubric.

3.1 Motivation

DataDeception exists because the bottleneck in commercial-claims accountability is data-availability, not methodology: the rules of fair claims are well established (FTC §5; comparative-advertising case law [1, 7]), but the evidence is scattered across paywalled magazines, ephemeral manufacturer URLs, and unstructured forum threads. The dataset is intended to support three audiences—**consumers** (a citable record of where a specific product diverges from claim), **researchers** (cross-product analysis at the tactic level), and **regulators** (a falsifiable corpus that can be challenged record-by-record). Construction is governed by two methodological commitments: mandatory citation on both claim and reality sides, and explicit handling of categorical (non-numeric) deceptions for which a numeric gap does not exist.

3.2 Composition: source taxonomy

Sources are tagged with a tier:

- **Tier 1 – Regulatory / open-government.** EPA fueleconomy.gov [23], NHTSA recalls.gov [15], FTC press releases, court rulings. Authoritative because produced under legal obligation with published methodology.

- **Tier 2 – Independent measurement.** Consumer Reports, MotorTrend, Edmunds, InsideEVs, Out of Spec Reviews, Recharged synthesis. Methodology published; test units typically purchased anonymously.
- **Tier 3 – Community / forum.** Owner forums (Tesla Motors Club, F-150 Lightning Forum, Rivian Forums), high-engagement Reddit threads. Cited only when $N \geq 10$ corroborating reports.
- **Tier 4 – Manufacturer / brand.** Treated as the *claim*, not the truth. Establishes what the manufacturer is asserting.

Figure 2 illustrates the actual source-tier composition of records in the v0.1 dataset.

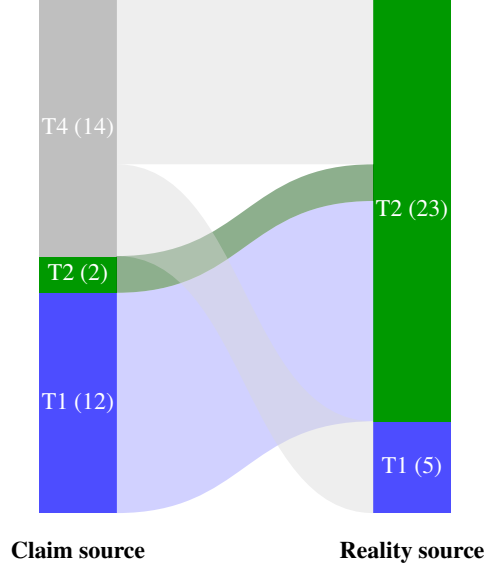


Figure 2: Source-tier composition of v0.1 records ($n = 28$). Left: claim sources (predominantly Tier 4 manufacturer surfaces and Tier 1 EPA filings). Right: reality sources (predominantly Tier 2 independent measurement).

3.3 Collection: evidence-grade rubric

A manufacturer assertion (typically Tier 4, the marketing surface) and a regulatory filing (typically Tier 1, e.g. EPA fueleconomy.gov) are paired with one or more independent measurements (Tier 2; or Tier 3 with $N \geq 10$). Each record carries an evidence grade $g \in \{A, B, C, D\}$:

- **A – Strong evidence.** Claim from Tier 1/4 with verbatim quote, *and* reality from Tier 1, or ≥ 2 converging Tier 2 measurements, or one Tier 2 plus Tier 3 with $N \geq 10$.
- **B – Moderate evidence.** Single Tier-2 source with documented methodology; or editorial framing of a manufacturer-disclosed number; or Tier-3 with $N \geq 5$.
- **C – Weak.** Single Tier-3 source, smaller N , or older than 36 months.
- **D – Anecdotal/preliminary.** Tier-4 only, single anecdotal post, or theoretical inference.

A-graded records carry the implicit claim that an informed reader can reproduce the finding from the cited sources alone. The rubric is in the spirit of Raji et al.’s [19] internal-audit instruments but adapted to the consumer-products domain: the evidentiary bar tracks the legal threshold for a deceptive-marketing finding rather than the algorithmic-fairness threshold.

3.4 Preprocessing: gap formalism and schema validation

Gaps are explicitly typed:

- **Numeric:** $\Delta_{\text{abs}} = r - c$, $\Delta_{\text{pct}} = 100 \cdot (r - c)/c$ where c and r are claim and reality values. Negative means reality is below claim.

- **Categorical:** naming/classification mismatches (e.g., “Full Self-Driving” vs. SAE Level 2). No Δ ; severity is required.
- **Definitional:** claimed and measured properties are different things despite shared naming (e.g., “motion rate” vs. refresh rate). No Δ ; severity is required.

Severity for non-numeric gaps is one of `critical` / `high` / `moderate` / `low` / `informational`, with `critical` reserved for cases with active regulatory finding, judicial ruling, or documented safety-of-life impact. Each completed record is emitted as a JSON file and validated against the schema in §4.1; the entire v0.1 collection passes validation under `validate.py`.

3.5 Uses

The dataset is designed to answer five research questions: (RQ1) What is the magnitude and direction of claim/reality divergence in EVs at the spec level? (RQ2) Which deception tactics recur across manufacturers, and which are idiosyncratic? (RQ3) How does Tier-1 regulatory data (NHTSA recalls) correlate with Tier-4 brand quality assertions? (RQ4) Where do counter-examples lie? (RQ5) What does the per-record citation chain reveal about source ecology over time? Pre-registered thresholds (e.g., ≥ 3 manufacturers for tactic-level corroboration) are documented in `RESEARCH_QUESTIONS.md` in the repository. Out-of-scope uses include automated scraping for litigation discovery (records are research artifacts, not pleadings) and any application that treats the v0.1 sample as representative of the full US EV market.

3.6 Distribution

The dataset, schema, validators, and figure-generation pipeline are released under MIT (code) and CC-BY-4.0 (data) at <https://github.com/Princeu3/DataDeception>. Each record cites manufacturer URLs via `web.archive.org` snapshots; this both addresses link rot and makes the corpus citable in the standard academic sense. A consumer-facing surface (`CONSUMER_REPORT.md`) is auto-generated from the same JSON.

3.7 Maintenance

Records carry a `last_verified` timestamp; software-driven products require re-verification at ≥ 12 month intervals because OTA updates can materially alter range, charging, and feature behavior. The repository maintains a versioned schema (`schema/`) and a `CHANGELOG` of taxonomy revisions. Manufacturers receive a notification of the relevant records concurrent with public release of v0.1; responses returned within a 30-day post-publication window are integrated as additional sources rather than as redactions, preserving the audit trail.

4 Schema and Tactic Taxonomy

4.1 JSON Schema

The dataset is structured as four kinds of JSON files (`products/`, `deceptions/`, `tactics/`, `schema/`). Every file validates against JSON Schema 2020-12, with referential-integrity checks: every `deception.product_id` resolves to an existing product file, every `deception.tactic_ids[]` resolves to a tactic file, and identifier conventions are mechanically enforced (kebab-case for products, snake_case for tactics). A reference validator (`validate.py`) ships with the dataset; the entire v0.1 collection passes validation.

4.2 Tactic taxonomy

A *tactic* is a repeatable pattern by which a manufacturer claim diverges from reality across multiple products. Tactics are the unit of theoretical interest in our framework: individual records are observations of tactics, and analysis at the tactic level supports cross-cutting research questions (does a given tactic recur across manufacturers? See §5.3).

The v0.1 taxonomy has 13 tactics across 5 families (Figure 3). The taxonomy is open: a new tactic may be added when ≥ 2 independent products exhibit a pattern not covered by existing tactics. Patterns observed in only one product are noted in the relevant record’s narrative but are not promoted to the dictionary until corroborated.

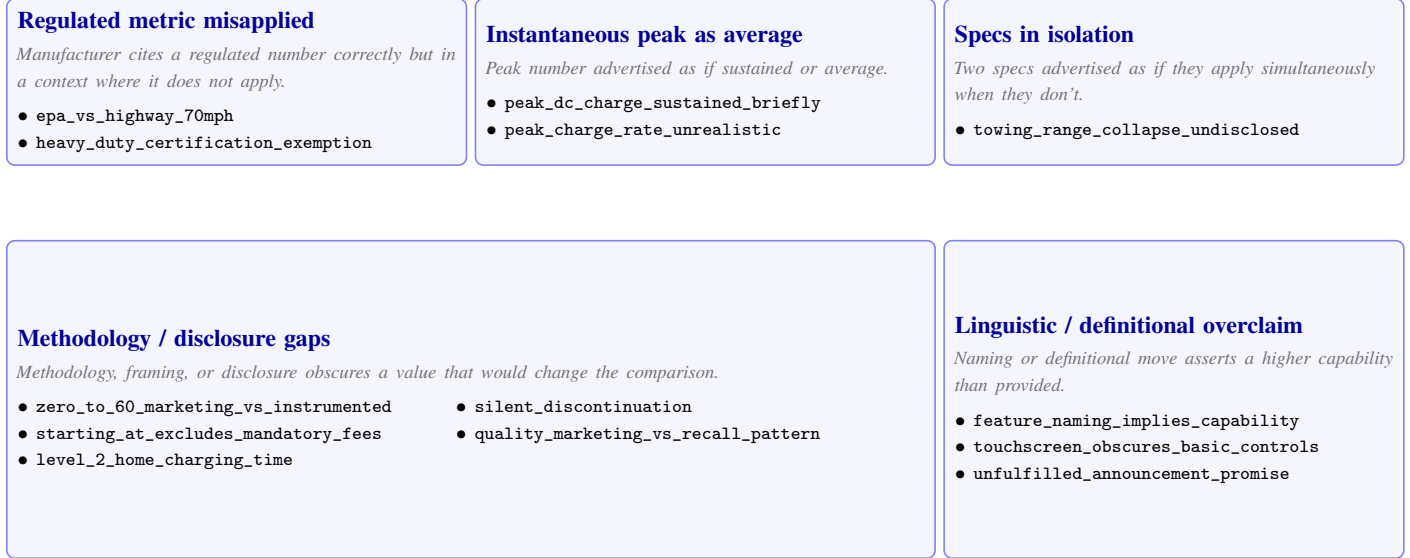


Figure 3: Tactic taxonomy v0.1. Five high-level families decompose into 13 named tactics. Each is referenced from one or more deception records; cross-product recurrence is the principal evidence for promoting a single-product observation into the taxonomy. Rendered as a 3-column card grid to fit a portrait page without scaling.

5 Preliminary Findings

We instantiate the methodology with 13 EV products (Tesla Model Y / Model 3 / Cybertruck; Ford F-150 Lightning / Mustang Mach-E; Rivian R1S; Hyundai Ioniq 5 / Ioniq 6; Mercedes EQS 450+; GMC Hummer EV; VW ID.4; Lucid Air Grand Touring; Polestar 2). Selection is non-randomized and deliberately includes products with publicly-documented gaps as well as counter-examples (Mercedes EQS, Ford Mach-E) where independent measurement matches or exceeds EPA. The 28 deception records cover 8 axes: range, charging, performance, software_features, pricing, production_status, ui_ergonomics, build_warranty.

5.1 Range divergence

Across products with documented `range__highway_70mph` records, mean gap is -15% with a range of -30% (Tesla Cybertruck [22]) to nearly 0 (Ford Mach-E, Mercedes EQS [6]). Figure 4 shows the per-product distribution.

The Tesla Cybertruck is the largest documented gap in our dataset and is given a longitudinal treatment in §5.6. In the opposite direction, Mercedes EQS represents the counter-example: Edmunds’ EV range loop measured 422 mi against a 350 mi EPA estimate [6], exceeding by 72 mi. We include this case to mitigate confirmation bias: the dataset is constructed to reflect both directions of error.

5.2 Categorical deception: feature-naming overclaim

The most-cited categorical deception is Tesla’s “Full Self-Driving”. A California administrative law judge ruled in December 2025 that the name was “unambiguously false and counterfactual” [16]; NHTSA opened a 2.88-million-vehicle investigation in October 2025 connecting FSD operation to 14 crashes and 23 injuries. Tesla complied with the required name change (“Full Self-Driving (Supervised)”, \$99/month subscription) but is suing the DMV to vacate the false-advertiser label [10]. Severity: *critical*.

5.3 Cross-manufacturer tactic recurrence (RQ2 evidence)

A central research question is whether deceptions are *idiosyncratic* (a feature of specific manufacturers) or *patterned* (a feature of industry-wide convention). Our preliminary evidence (Figure 5) supports the latter: the `epa_vs_highway_70mph` tactic appears in records spanning 8 distinct manufacturers (Tesla, Ford, Rivian, Hyundai,

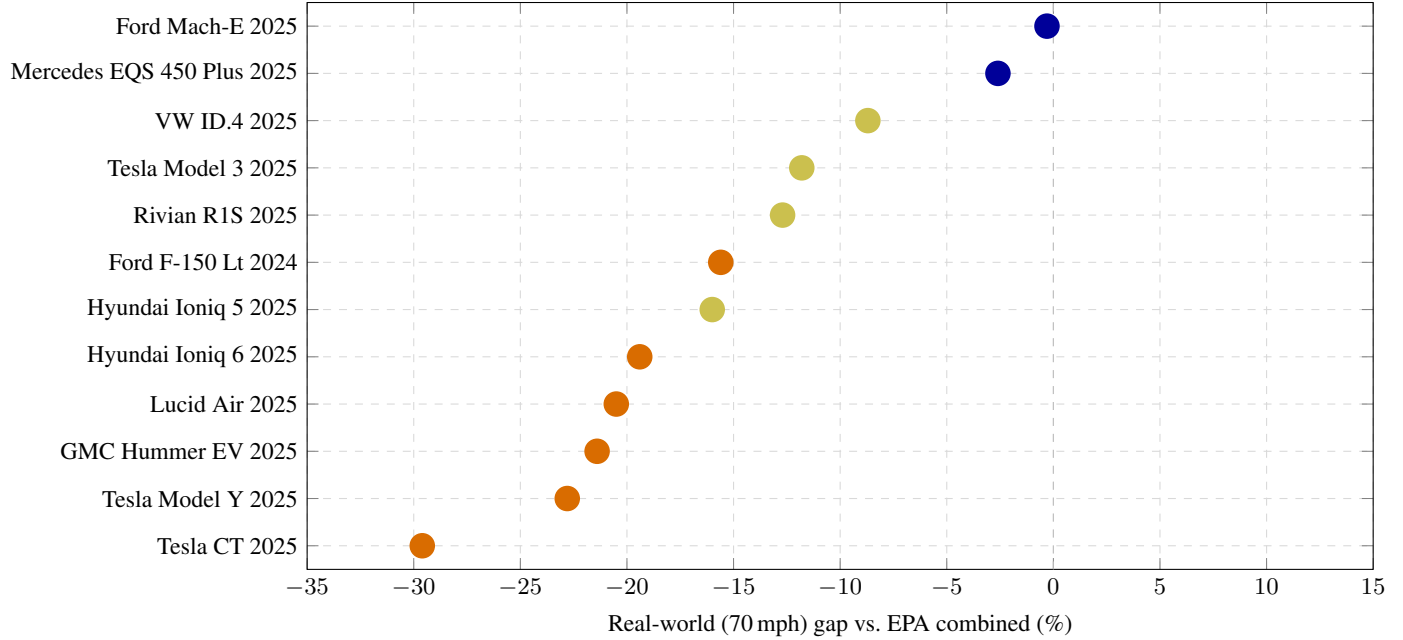


Figure 4: Per-product highway-range gap (real-world 70 mph vs. EPA combined). Markers colored by severity (red: critical/high, yellow: moderate, blue: counter-example).

Lucid, Mercedes-Benz, GMC, Volkswagen) including counter-examples in both directions, decisively exceeding our pre-registered threshold of ≥ 3 for tactic-level cross-cutting evidence. The `peak_dc_charge_sustained_briefly` and `quality_marketing_vs_recall_pattern` tactics each appear in 3 manufacturers, also clearing the threshold.

The patterning has a clear structural cause. The dominant tactic is not a coincidence of marketing choices: it is a consequence of EPA’s combined city/highway test cycle being the only mandatory range figure manufacturers must publish, while the customer-relevant condition (sustained 70 mph cruising) is not. Every manufacturer faces the same incentive to publish the higher EPA number; few publish the supplemental 70 mph figure independent measurement reveals. The cross-cutting structure is therefore evidence of *regulatory test-cycle dominance*, not of coordinated marketing intent. This is an important distinction for the policy framing in §5.7: the appropriate intervention is at the disclosure layer (mandate the second figure) rather than at the deceptive-intent layer (require findings of culpability), which would face a much higher evidentiary bar.

5.4 Severity composition across axes

Figure 6 shows the per-product, per-axis severity grid. Range gaps are nearly universal but typically *moderate* or *high*. The *critical* cells are concentrated in two product/axis pairs: Tesla Model Y / FSD (regulatory ruling) and Ford F-150 Lightning / towing (stranding-risk safety angle). Counter-example cells (informational severity, blue) are present.

Two structural observations follow from the grid. First, severity is not uniformly distributed across axes: range and recall axes are widely populated (consistent with their independent-measurement infrastructure being mature), while pricing, software, and `ui_ergonomics` axes are sparser (consistent with the absence of standardized comparison instruments for those dimensions). Empty cells should therefore not be read as absence of deception but as absence of measurement; this is one of the limitations addressed in §6. Second, the two *critical* cells share a common structural feature: each has both a regulatory finding (NHTSA / California DMV) and an independent-measurement record (CR / MotorTrend). Critical severity in our rubric is reserved for cases where these two evidence streams converge, which is intentionally rare and intentionally robust.

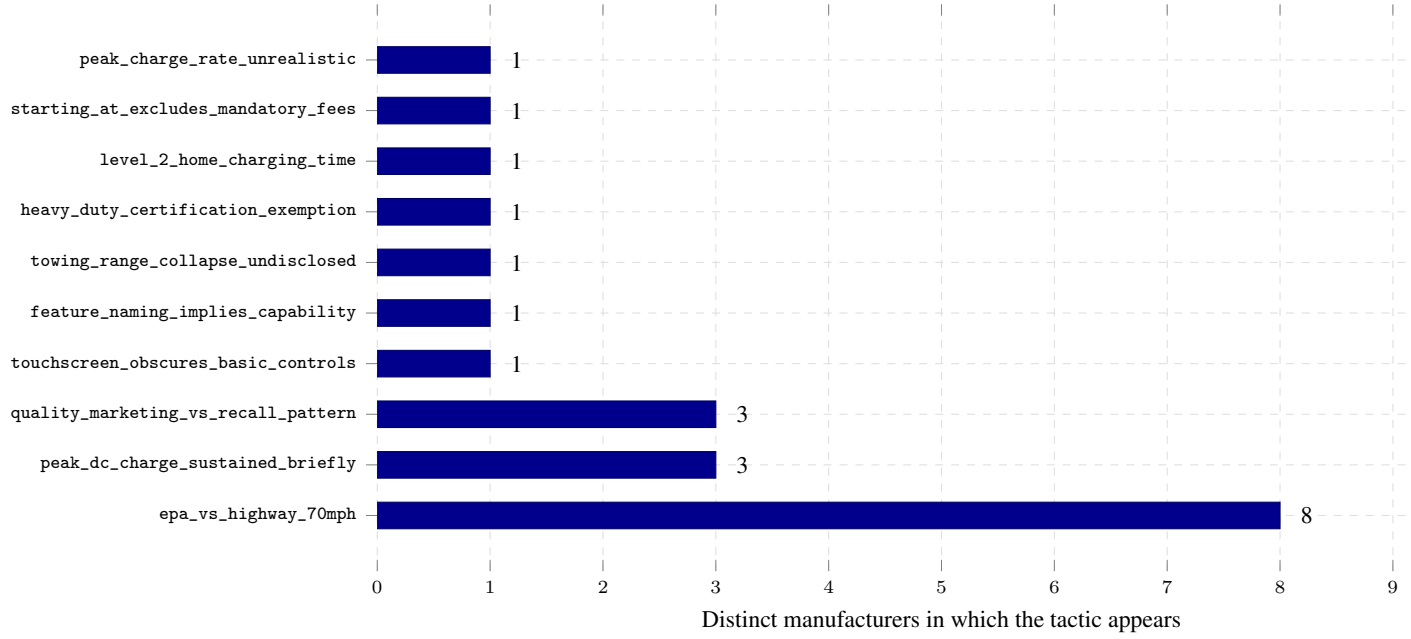


Figure 5: Top tactics by number of distinct manufacturers in which they appear. `epa_vs_highway_70mph` dominates: a tactic spanning 8 manufacturers cannot be characterized as idiosyncratic. The next two tactics (`quality_marketing_vs_recall_pattern` and `peak_dc_charge_sustained_briefly`) each span 3 manufacturers, also clearing the pre-registered ≥ 3 threshold.

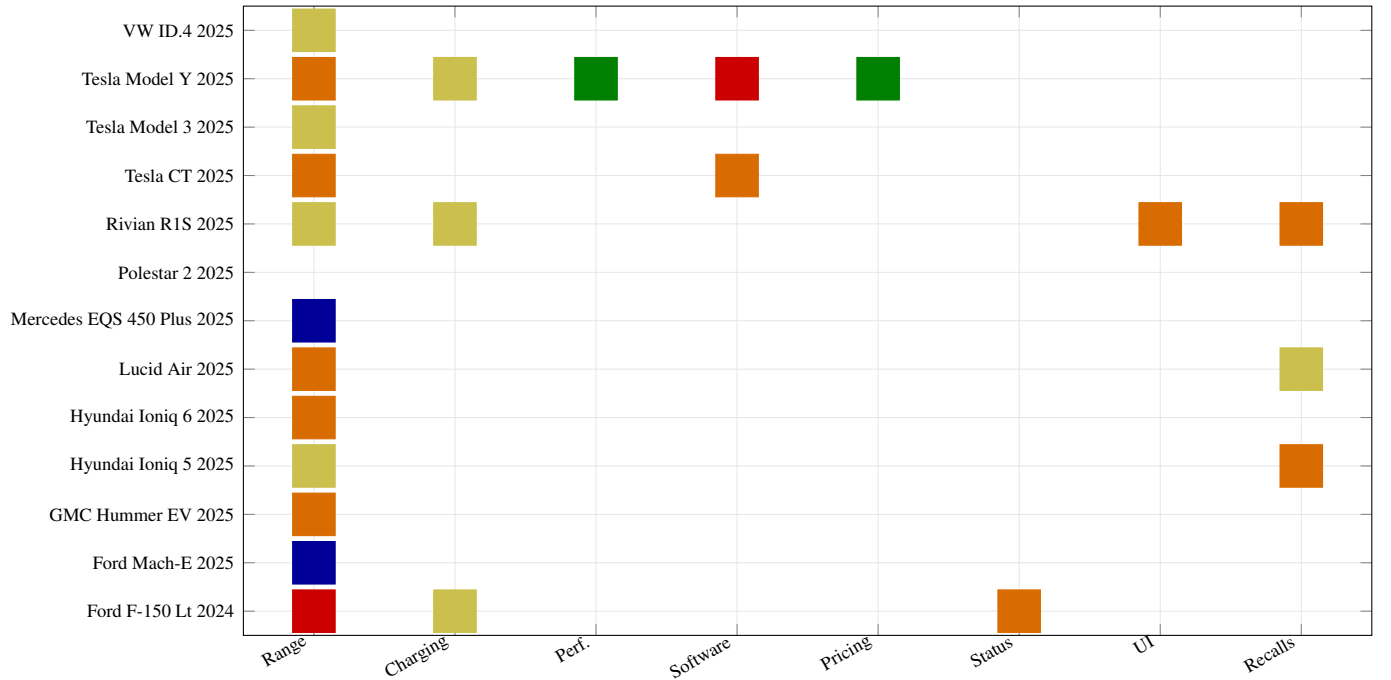


Figure 6: Severity heatmap: products $\times$ deception axes, colored by maximum severity at that intersection. Empty cells indicate no deception record (not necessarily absence of deception).

5.5 Tier-1 recall density (build/warranty axis)

NHTSA recall counts for the documented model year (Figure 7) reveal substantial inter-manufacturer variance. Rivian R1S 2025 (9 recalls), Hyundai Ioniq 5 2025 (8, including two separate high-voltage traction-battery campaigns), and Lucid Air 2025 (5) are notable. Several products show 0 recalls for the documented model year (Ford Mach-E, F-150 Lightning, Mercedes EQS, GMC Hummer EV, Polestar 2). The brand assurance language used by manufacturers does not, in general, reflect this variance.

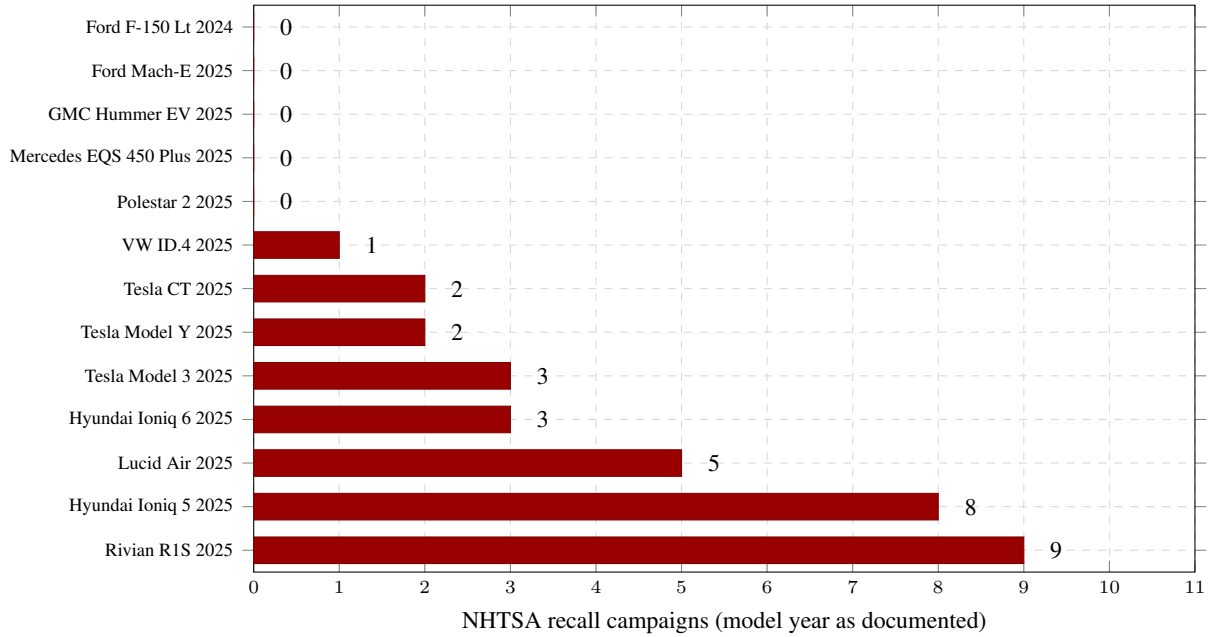


Figure 7: NHTSA recalls for the documented model year, by product. Rivian R1S, Hyundai Ioniq 5, and Lucid Air dominate; quality-marketing language does not, in general, reflect first-year defect-discovery rate.

5.6 Case study: Tesla Cybertruck range promise (2019–2025)

A single product traces the full life of a numeric overclaim through marketing, regulatory testing, independent measurement, and a cancelled mitigation, providing a longitudinal view that aggregate findings cannot. We use the Tesla Cybertruck as our exemplar because the available evidence chain is complete and Tier-1/Tier-2 corroborated.

2019 – the original promise. At the November 2019 Cybertruck announcement, Tesla advertised three trims with maximum ranges of 250, 300, and “500+ miles” [20]. Reservations opened the same day; the 500-mile trim was the marketing centerpiece for a vehicle whose principal purchase justification, against an internal-combustion F-150, depended on parity with a tankful of gasoline.

2023 – launch and shipped trims. At delivery, no trim achieved the 500-mile promise on the EPA combined cycle. The shipped Dual Motor variant was rated 318 miles by EPA. To address the gap, Tesla announced a Range Extender accessory: an auxiliary battery occupying a third of the bed, sold for \$16,000, that would purportedly bring effective range above the 500-mile mark.

2024 – independent measurement. MotorTrend’s instrumented Road-Trip Range test [22] measured the Dual Motor at 224 mi sustained at 70 mph against the 318 mi EPA-rated figure, a 30% shortfall. MotorTrend characterized the result as “among the worst performers in our database of 56 EVs.” The Cybertruck thus sat at −30% on our `epa_vs_highway_70mph` tactic, the largest documented gap in the v0.1 corpus.

May 2025 – the cancellation. Tesla quietly cancelled the Range Extender, refunded all deposits, and removed the product from the configurator [20]. Six years after the original 500-mile promise, no buyer had received the accessory; the engineering challenge of meeting the original advertised range was effectively conceded.

What the case study illustrates. The Cybertruck record exhibits four of our five tactic families simultaneously: *specs in isolation* (the 500-mile figure not paired with battery weight or accessory disclosure), *regulated metric misapplied* (EPA combined-cycle figure used in marketing where buyers expected highway-cruise behavior), *methodology / disclosure gaps* (the Range Extender announcement absorbed accountability for the 500-mile figure but was never delivered), and *linguistic / definitional overclaim* (“500+ miles” as a categorical commitment that regressed to a conditional one). It is the cleanest illustration in our corpus that single-product claim/reality records, when constructed with full citation chains, become legible as case histories rather than isolated assertions.

5.7 Implications for consumer protection policy

Three implications follow directly from the v0.1 corpus and bear on near-term regulatory action.

1. Modernize the EPA test cycle for highway dominance. The dominant single tactic in the corpus (`epa_vs_highway_70mph`, recurring across 8 manufacturers) reflects a structural mismatch: EPA’s combined city/highway cycle weights low-speed urban driving more heavily than the use case (long-distance highway travel) for which range matters most to buyers. A single supplemental “70 mph constant-speed” rating, published on the same Monroney label, would force the dominant tactic into the regulatory frame and provide a directly-comparable second figure across manufacturers. This is administrable using EPA’s existing dynamometer infrastructure.

2. Treat feature-naming overclaim as a per-se UDAP violation when SAE-level mismatch is structural. Tesla’s “Full Self-Driving” was ruled “unambiguously false and counterfactual” by a California ALJ in December 2025 [16] after the name had been in market for years. The ruling pattern (administrative finding → litigation → partial naming concession → continued sale under a parenthetical disclaimer) establishes that the existing UDAP framework can reach categorical naming overclaim, but the lag is measured in years and a single-jurisdiction ruling is not nationally binding. A bright-line rule treating consumer-facing feature names that imply higher SAE autonomy levels than the system delivers as per-se deceptive (in the manner of “misbranding” for food and drugs) would compress the lag.

3. Mandate independent-test-result disclosure adjacent to manufacturer claims. The structural data-availability problem this paper is responding to—paywalled magazines, ephemeral manufacturer URLs, fragmented forum data [21]—suggests that voluntary independent measurement is not a stable accountability mechanism over time. A disclosure rule requiring that the manufacturer’s window-sticker range claim be accompanied by the most recent independent measurement (where one exists, with attribution to a registered measurement organization) would not require new measurement infrastructure but would require manufacturers to acknowledge it. The DataDeception schema is one candidate format for such an attached disclosure.

These implications are intentionally narrow: each maps to a single change in an existing legal or regulatory instrument rather than calling for new statutory authority. Building on Luguri and Strahilevitz’s [11] bridge between empirical dark-pattern audits and FTC §5 enforcement, we treat structured deception corpora as a precondition for, rather than a substitute for, regulatory action.

6 Limitations and Threats to Validity

This dataset is v0.1 and the sample is small ($n = 13$ products). We explicitly do not claim the findings generalize to the full US EV market; that requires a randomized expansion (Phase 2). Specific limitations:

- 1. Selection bias:** Products were chosen partly because Consumer Reports had documented large gaps for them. Phase 2 will randomize within the top-15 US-sales list.
- 2. Single-unit Tier-2 testing:** CR, Edmunds, MotorTrend tests typically use $N = 1$ purchased unit. We mitigate by triangulating across publishers when possible.
- 3. Software-update drift:** EVs receive OTA updates that materially affect range/charging behavior. Records timestamp `last_verified`; re-verification is required at ≥ 12 month intervals.
- 4. Manufacturer perspective absent:** No manufacturer was contacted for response. The dataset is, in v0.1 form, one-sided. A pre-publication notification template for ≥ 30 -day manufacturer response is included in the repository.
- 5. Source-reproducibility risk:** Some Tier-2 sources (CR) are partly paywalled; some Tier-4 manufacturer URLs change without warning. We mitigate via verbatim quotes and web.archive.org snapshots, but link rot is real.

6. **Inter-rater reliability not yet measured:** Tactic assignments were performed by a single annotator. Cohen’s κ on a sample of 20 records is a Phase 2 deliverable.

7 Conclusion and Future Work

DataDeception is a methodology-and-dataset for systematically documenting commercial-claims deception, instantiated for the U.S. electric-vehicle market. The contribution is primarily the *method*: a citation-first schema with mandatory tier-classified sources, an open tactic taxonomy with cross-manufacturer corroboration thresholds, and an evidence-grade rubric that admits replication. The v0.1 dataset is small but supports preliminary RQ1 and RQ2 testing and demonstrates the feasibility of a public, schema-validated artifact in this space.

Phase 2 work prioritizes: randomized expansion to 25+ products, longitudinal re-verification, manufacturer-response integration, inter-rater reliability measurement, and instantaneous-snapshot pipeline for Tier-4 manufacturer URLs. The dataset and methodology generalize beyond EVs; we anticipate parallel instantiations for televisions (subject to RTINGS access constraints), laptops (Notebookcheck), and audio (Audio Science Review). The same schema applies; only the source-tier-2 base differs.

Reproducibility statement. All artifacts are available at <https://github.com/Princeu3/DataDeception> under MIT (code) and CC-BY-4.0 (data). The figure-generation pipeline is deterministic and reads only the JSON dataset; running `paper/figures/generate.py` reproduces all auto-derived figures. Schema validation is via `validate.py` (one command, exits non-zero on any integrity violation).

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